

Research Interests

- Climate change effects on hydrological extremes
- Dynamics of mesoscale convective systems
- Regional and global km-scale modeling
- Satellite observations of clouds and precipitation

Research positions

- May 2026–present **AFESP Research Fellow**, *Department for Meteorology, University of Reading.*
- Oct 2023–April 2026 **ASP Postdoctoral Fellow**, *Mesoscale and Microscale Meteorology Lab, NSF National Center for Atmospheric Research, Boulder, Colorado.*
- Mar 2023–Sep 2023 **Research Assistant in Mesoscale Meteorology**, *Regional Climate Group, Department of Earth Sciences, University of Gothenburg, Sweden.*

Education

- Sep 2018–Mar 2023 **Doctor of Philosophy (Ph.D.) in Natural Sciences**, *Regional Climate group, Department of Earth Sciences, University of Gothenburg, Sweden,*
Thesis project: Observing and Modeling Precipitation in the Tibetan Plateau region.
Advisor: Prof. Deliang Chen
- Sep 2016–May 2018 **Master of Science in Atmospheric Sciences**, *Final grade: VG (Excellent),*
Department of Earth Sciences, University of Gothenburg, Sweden,
Thesis project: Temporal and spatial variability of convection, clouds and precipitation over the Tibetan Plateau derived from recent satellite retrievals.
Advisor: Prof. Deliang Chen
- Sep 2013–May 2016 **Bachelor of Science in Earth Sciences with Major in Climatology**, *Final grade: VG (Excellent),*
Department of Earth Sciences, University of Gothenburg, Sweden,
Thesis project: Major ion deposition in the accumulated winter snowpack in northern Sweden.
Advisor: Prof. Hans Linderholm

Research visits

- Apr–Sep 2023 **National Center for Atmospheric Research, Boulder, Colorado, USA**,
Visitor in the Climate Extremes Section, Host: Prof. Andreas Franz Prein and Dr. James Done,
Project: Evaluation of convective storms in kilometer-scale climate simulations.
- Oct 2021–May 2022 **National Center for Atmospheric Research, Boulder, Colorado, USA**,
ASP Graduate visitor program, Host: Prof. Andreas Franz Prein,
Project: Ensemble-based convection-permitting simulations in the Third Pole region.
- Oct–Dec 2017 **School of Atmospheric Sciences, Nanjing University, China**,
Research visit in Aerosol-cloud research group, Host: Prof. Minghuai Wang,
Project: Satellite observations of convective clouds over the Tibetan Plateau.
- Jun–Aug 2016 **Max Planck Institute for Meteorology, Hamburg, Germany**,
Internship in Hydrological group, Host: Dr. Tobias Stacke,
Project: Validation of a global dynamical wetland scheme in land-atmosphere coupled simulations.
- Jun–Aug 2014 **Helmholtz Centre for Ocean Research, Kiel, Germany**,
Internship in Paleoclimatology and Natural Resources, Host: Dr. rer. nat. Warner Brückmann.

Invited talks

- July 2026 **University of Reading,**
Host: Prof. Thorwald Stein, ParaChute conference,
Title: Precipitation efficiency in the convective gray zone: grid-spacing dependence, observational constraints and storm-type variability .
- April 2026 **University of Vienna,**
Host: Prof. Blaz Gasperini,
Title: Satellite-retrieved Cloud Ice as a Constraint on Precipitation Efficiency in Convective Storms.
- Apr 2025 **Duke University, Durham, USA.,**
Host: Prof. Wenhong Li, Nicholas School of the Environment,
Title: Mesoscale convective systems and Precipitation in a Changing Climate.
- Mar 2024 **Pacific Northwest Laboratory, Washington, USA,**
Host: Dr. Zhe Feng, Atmospheric Sciences and Global Change Division,
Title: Mesoscale convective systems and their Impact on Precipitation.
- Nov 2023 **Leipzig University, Germany,**
Host: Dr. Matthias Tesche, Leipzig Institute for Meteorology,
Invited lecture for *Advanced Training Module (ATM) on Remote Sensing of Cloud Development within the framework of the Leipzig Graduate School on Clouds, Aerosols and Radiation (LGS-CAR):*
Development of Deep Convection and how we can use Storm Tracking to study it.

Fellowships and Grants

- 2025-2026 **Strategic Funding Grant for NCAR's Mesoscale & Microscale Meteorology Laboratory,**
Project: Feature-Based Assessment of Extreme Precipitation in global and regional MPAS simulations,
Colorado, USA.
- 2024-2026 **ASP Postdoctoral Fellowship,**
Project: How well do km-scale climate models represent organized convection beyond surface precipitation?,
Colorado, USA.
- 2021 **NCAR Advanced Study Program for graduate visitors,**
Colorado, USA.
- 2019 **Travel fund to International Conference on Regional Climate-CORDEX 2019,**
Beijing, China.
- 2018 **Research Fund Adlerbertska Stiftelse,**
Sweden.
- 2018 **Sven Lindqvists forskningsstiftelse för doktorandstudier vid Geovetarcentrum,**
Gothenburg, Sweden.
- 2018-2023 **Swedish National Space Agency (SNSA) Grant,**
Project: The role of convection for precipitation in the Third Pole region - Satellite and ground-based observation vs model simulations, Grant nr. 188/18 4).

Awards and Honors

- 2023 **Faculty of Science Doctoral Thesis Award at the University of Gothenburg,**
Gothenburg, Sweden.
- 2021 **Wiley Top downloaded article,**
Title: *The Role of Mesoscale Convective Systems in Precipitation in the Tibetan Plateau region.*

Outreach and Engagement

- 2018–2021 **Coordinator and Steering Board member of the Gothenburg Air and Climate Network (GAC).**
- 2018–2021 **Executive Secretary of the Association of Polar and Alpine Early Career Scientists (APECS).**
- 2024–present **Group member of the NCAR Water system program (including CONUS404 and the South America Affinity Group).**
- 2024–present **Steering board member of the NCAR Early Career Scientist Assembly (ECSA).**

- 2025–present **Member of the WCRP Global Precipitation Experiment (GPEX), Working group 3 (Prediction and Modeling).**
- 2026–present **Member of the WCRP Digital Earth Km-scale Working Group.**
- 2026–present **Chair of the GEWEX Convective tracking working group on Scale Interactions.**

Contributions to research community

Reviewer for the following scientific journals,

JGR Atmosphere, Journal of Climate, Journal of Applied Meteorology and Climatology, International Journal of Climatology, Geoscientific Model Development, Geoscience Data Journal, Climatic Change.

Co-convener for sessions at the following scientific conferences,

European Geoscience Union: Advancing understanding of the circulation-coupling and Lagrangian evolution of clouds

American Meteorological Society: Climate Impacts of Tropical Ice Clouds .

Pedagogical courses

- 2025 **Early Career Leadership Program,**
NSF NCAR, Boulder, Colorado.
- 2019 **Teaching and Learning in Higher Education as a Scientific Field,**
University of Gothenburg, Sweden.

Supervision and Mentoring

- 2025 **Shanna Chamhitt (Graduate student in NCAR's SOARS program),**
Project: Using satellite observations to measure non-propagating gravity waves in tropical cyclones, NSF NCAR, Boulder, Colorado.
- 2022 **Benjamin Odenman Holmberg (Master thesis),**
Project: Analysis of mesoscale weather systems in the Tibetan plateau region at different resolution using ERA5, HAR and WRF,
University of Gothenburg, Sweden.
- 2022 **Anna Dugoul (Master thesis),**
Project: Characterisation of extreme precipitation events in the Sichuan Basin based on in-situ observations and their connection to mesoscale convective systems and other mesoscale disturbances,
University of Gothenburg, Sweden.
- 2021 **Devashree Niraula (Master thesis),**
Project: Assessing the role of extreme precipitation on the Babai river basin and its impact on the Bhari Babai Mulidiversion Project,
University of Gothenburg, Sweden.
- 2019 **Raquel Flügel (Bachelor thesis),**
Project: Evaluating Modelled Precipitation against ground-based Observation and Satellite Data above the Tibetan Plateau,
University of Gothenburg, Sweden.

Technical skills

Computer	Python (<i>Advanced</i>), Linux and Bash scripting (<i>Good</i>), NCO/CDO (<i>Good</i>), R (<i>Basic</i>), Matlab (<i>Basic</i>)
Utilities	Anaconda, Git, Jupyter Notebook, Slurm
Numerical Models	Weather Research and Forecasting (WRF) Model, Model Prediction Across Scales (MPAS)
Languages	German (<i>Mothertongue</i>), English (<i>Fluent</i>), Swedish (<i>Fluent</i>), French (<i>Good</i>), Spanish (<i>Basic</i>)